

Olympiad Test Paper — Mathematics (Class 7) — Paper 3

Chapter 7: A Tale of Three Intersecting Lines

Paper 3 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	7 — A Tale of Three Intersecting Lines
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Classic triangle traps: equal-sum sides fail the inequality (a sum must EXCEED, not merely match, the third side), an exterior angle equals the SUM of the two remote interior angles (not the adjacent one), 'acute' demands ALL three angles below 90° , an altitude is a perpendicular (not a side or median), and the largest angle always sits opposite the longest side. Do not write on this paper — mark answers on your answer sheet.

1. In triangle ABC the three angles are in the ratio 3 : 5 : 7. Side BC is extended at the vertex with the LARGEST interior angle to form an exterior angle there. What is that exterior angle?

- (A) 84° (B) 96°
(C) 108° (D) 72°

2. In triangle PQR, side QR is extended beyond R to S so that the exterior angle $\angle PRS$ measures 124° . Inside the triangle $\angle Q$ is 26° smaller than $\angle P$. Find $\angle P$.

- (A) 49° (B) 62°
(C) 75° (D) 98°

3. Two sides of a triangle measure 13 cm and 6 cm and the third side is a whole number of centimetres. How many different whole-number values can the third side take?

- (A) 12 (B) 13
(C) 11 (D) 7

4. In an isosceles triangle the two equal sides are AB and AC, and the vertex angle $\angle A$ is 44° . An angle bisector is drawn from A to BC. What is the measure of EACH base angle of

the triangle?

- (A) 44°
- (B) 88°
- (C) 46°
- (D) 68°

5. In triangle ABC, $\angle A = 2x + 10^\circ$, $\angle B = 3x - 20^\circ$ and $\angle C = x + 40^\circ$. Classified by its angles, the triangle is:

- (A) Acute (all angles below 90°)
- (B) Right-angled
- (C) Obtuse
- (D) Equiangular (all angles equal)

6. A triangle has sides 9 cm, 9 cm and 9 cm. A second triangle has angles 100° , 40° and 40° . Which single statement correctly classifies BOTH?

- (A) The first is equilateral and right; the second is isosceles and acute
- (B) The first is equilateral and acute; the second is isosceles and obtuse
- (C) Both are isosceles and acute
- (D) The first is scalene; the second is right-angled

7. In triangle ABC the altitude from A meets BC at its foot D, and the median from A meets BC at its midpoint M. If these two points D and M are the SAME point, what must be true of the triangle?

- (A) The triangle must be equilateral
- (B) The triangle must be right-angled at A
- (C) BC is the longest side
- (D) $AB = AC$ (the triangle is isosceles with apex A)

8. In triangle ABC, $AB = 7$ cm, $BC = 9$ cm and $CA = 5$ cm. Without measuring, which angle is the LARGEST?

- (A) $\angle B$ (opposite CA)
- (B) $\angle C$ (opposite AB)
- (C) They are all equal
- (D) $\angle A$ (the angle opposite BC)

9. In triangle ABC, $\angle A$ exceeds $\angle B$ by 18° and $\angle C$ is twice $\angle B$. Find $\angle C$.

- (A) 40.5°
- (B) 81°
- (C) 60°
- (D) 63°

10. A triangle has two angles measuring 35° and 85° . Its sides are then arranged from shortest to longest. Which ordering is correct?

- (A) Side opposite 35° < side opposite 60° < side opposite 85°
- (B) Side opposite 85° < side opposite 60° < side opposite 35°
- (C) All three sides are equal
- (D) Side opposite 60° is the longest

11. In triangle ABC, side AB is extended beyond B to a point E. The exterior angle $\angle CBE = 3\angle C$ and $\angle A = 50^\circ$. Find $\angle C$.

- (A) 75°
- (B) 25°
- (C) 50°
- (D) 37.5°

12. Two sides of a triangle are 8 cm and 15 cm, and its perimeter is a whole number. The third side is the SMALLEST whole-number length possible. The triangle's perimeter is

then:

- | | |
|-----------|-----------|
| (A) 30 cm | (B) 31 cm |
| (C) 23 cm | (D) 38 cm |

13. In an isosceles triangle, each of the two EQUAL angles is 15° more than the third (unequal) angle. Each equal angle measures:

- | | |
|----------------|----------------|
| (A) 50° | (B) 70° |
| (C) 65° | (D) 55° |

14. In triangle ABC, $\angle A : \angle B = 2 : 3$ and $\angle C = 80^\circ$. Find $\angle B$.

- | | |
|-----------------|----------------|
| (A) 40° | (B) 50° |
| (C) 100° | (D) 60° |

15. A triangle has angles 90° , 60° and 30° . Considering the altitudes drawn from its three vertices, where do they all meet?

- (A) At the midpoint of the hypotenuse (B) Outside the triangle
(C) At the centre of the longest side
(D) At the vertex of the 90° angle (the right-angle vertex)

16. In triangle ABC, $AB = AC$ and the angle bisector from A meets BC at D. If $\angle B = 64^\circ$, what is $\angle BAD$?

- | | |
|----------------|----------------|
| (A) 52° | (B) 32° |
| (C) 26° | (D) 64° |

17. Three lengths are $2k$, $3k$ and $4k$ centimetres (k a positive number). To also be the sides of a valid triangle, this set:

- (A) Always forms a triangle, because $2k + 3k = 5k > 4k$
(B) Never forms a triangle
(C) Forms a triangle only when k is a whole number
(D) Forms a triangle only when $k > 1$

18. In triangle ABC, the exterior angle at A is 140° and the exterior angle at B is 110° . Find the interior angle at C.

- | | |
|-----------------|----------------|
| (A) 110° | (B) 60° |
| (C) 70° | (D) 40° |

19. A triangle has one angle of 96° . Which statement about the remaining two angles is certainly true?

- (A) One of them could be a right angle (B) They are equal to each other
(C) One could be obtuse
(D) Both are acute and together they measure 84°

20. Two sides of a triangle are 10 cm and 10 cm. The third side is a whole number. The number of whole-number values it can take is:

- (A) 20
(C) 19
- (B) 10
(D) 21

21. In triangle ABC, $\angle A = (x + 30)^\circ$, $\angle B = (2x)^\circ$ and $\angle C = (x - 10)^\circ$. Find the value of the LARGEST angle.

- (A) 70°
(C) 80°
- (B) 60°
(D) 30°

22. In triangle ABC, $AB = AC$ and an exterior angle at the base vertex B measures 115° . Find the vertex angle $\angle A$.

- (A) 65°
(C) 50°
- (B) 115°
(D) 80°

23. Four candidate angle-triples are: $(45^\circ, 45^\circ, 90^\circ)$, $(70^\circ, 70^\circ, 40^\circ)$, $(100^\circ, 50^\circ, 30^\circ)$, $(60^\circ, 60^\circ, 60^\circ)$. How many of these describe an ISOSCELES triangle (at least two equal angles)?

- (A) 2
(C) 4
- (B) 3
(D) 1

24. In triangle ABC, side BC is extended both ways: beyond C the exterior angle is 125° and the interior angle $\angle B = 65^\circ$. Find $\angle A$.

- (A) 60°
(C) 65°
- (B) 55°
(D) 115°

25. A triangle has sides of 12 cm and 7 cm. Which of the following could NOT be the length of the third side?

- (A) 5 cm
(C) 15 cm
- (B) 8 cm
(D) 18 cm

26. In triangle PQR, the median from P meets QR at its midpoint M, and PM is drawn. Which statement is always TRUE about this median?

- (A) It is perpendicular to QR
(B) It divides triangle PQR into two smaller triangles of equal area
(C) It bisects the angle at P
(D) It equals half of QR

27. Two sides of a triangle are 14 cm and 9 cm. The third side is a whole number that is also EVEN. How many such values are possible?

- (A) 8
(C) 17
- (B) 10
(D) 9

28. In triangle ABC, $\angle B = \angle C$ and the side opposite $\angle A$ is 6 cm while the side opposite $\angle B$ is 10 cm. What is the side opposite $\angle C$?

- (A) 6 cm
(C) 10 cm
- (B) 8 cm
(D) 16 cm

29. The three exterior angles of a triangle (one at each vertex) are $(2y)^\circ$, $(3y)^\circ$ and $(4y)^\circ$. The smallest INTERIOR angle of the triangle is:

- (A) 40° (B) 20°
(C) 80° (D) 60°

30. In triangle ABC, $\angle A = 90^\circ$. The altitude from A to the hypotenuse BC has its foot at H. Which statement is correct?

- (A) H coincides with B or C (B) AH lies outside the triangle
(C) H lies on BC, strictly between B and C, and AH is perpendicular to BC
(D) AH equals the hypotenuse BC

31. In triangle ABC, $\angle A = 40^\circ$. The bisectors of $\angle B$ and $\angle C$ meet at point I. What is $\angle BIC$?

- (A) 70° (B) 140°
(C) 90° (D) 110°

32. In triangle ABC, the angles satisfy $\angle A - \angle B = 20^\circ$ and $\angle B - \angle C = 20^\circ$. Find $\angle B$.

- (A) 60° (B) 40°
(C) 80° (D) 50°

33. A triangle has sides 5 cm, 5 cm and 6 cm. Considering both sides and angles, the BEST classification is:

- (A) Isosceles and acute (B) Isosceles and right-angled
(C) Equilateral and acute (D) Scalene and obtuse

34. In triangle ABC, $\angle A = 2\angle C$ and $\angle B = \angle A - 20^\circ$. Find $\angle C$.

- (A) 80° (B) 60°
(C) 40° (D) 50°

35. In triangle ABC, a line through A parallel to BC is drawn. If $\angle B = 55^\circ$ and $\angle C = 65^\circ$, the two angles this parallel line makes with AB and AC on the side of the triangle add up to:

- (A) 60° (B) 110°
(C) 130° (D) 120°

36. In triangle ABC, the median AM and the altitude AH are both drawn from A to side BC. If $\angle B = 40^\circ$ and $\angle C = 70^\circ$, which is true about points M and H?

- (A) M and H always coincide (B) H is the midpoint of BC
(C) M is the foot of the perpendicular
(D) M and H are different points (the triangle is not isosceles at A)

37. A triangle has two sides 16 cm and 9 cm. The third side is a whole number that makes the triangle ISOSCELES. How many such whole-number values are there?

- (A) 2 (B) 1
(C) 3 (D) 0

38. In triangle ABC, the exterior angle at C equals 150° and $\angle A = 4\angle B$. Find $\angle A$.

- (A) 30°
(C) 120°

- (B) 96°
(D) 100°

39. A triangle has perimeter 36 cm and its sides are in the ratio 3 : 4 : 5. The longest side, and the type of triangle by angles, are:

- (A) 15 cm; right-angled
(C) 12 cm; right-angled

- (B) 15 cm; acute
(D) 20 cm; obtuse

40. In triangle ABC, $\angle A = 70^\circ$. The two angles $\angle B$ and $\angle C$ are such that $\angle B$ is 3 times $\angle C$. Find $\angle B$.

- (A) 82.5°
(C) 55°

- (B) 27.5°
(D) 110°

41. In an isosceles triangle the vertex angle is 4 times each base angle. The vertex angle measures:

- (A) 30°
(C) 120°

- (B) 60°
(D) 90°

42. Two sides of a triangle are 20 cm and 13 cm. The third side x is a whole number with $13 \leq x \leq 20$. How many such values give a valid triangle?

- (A) 7
(C) 33

- (B) 6
(D) 8

43. In triangle ABC, $\angle A = 50^\circ$. The bisector of $\angle A$ meets BC at D, and $\angle ABD = 60^\circ$. Find $\angle ADB$.

- (A) 95°
(C) 70°

- (B) 85°
(D) 65°

44. A triangle is described as having all three angles different, the largest being 95° . Classified by sides and by angles, it is:

- (A) Scalene and obtuse
(C) Isosceles and obtuse

- (B) Scalene and acute
(D) Scalene and right-angled

45. In triangle ABC, the foot of the altitude from A lies OUTSIDE segment BC, on the extension beyond B. What does this tell you about the triangle?

- (A) The angle at A is obtuse
(C) The triangle is right-angled at B

- (B) The angle at B is obtuse
(D) The triangle is equilateral

46. Two sides of a triangle are 25 cm and 18 cm. The third side is a whole number. The difference between the LARGEST and SMALLEST possible third sides is:

- (A) 43
(C) 33

- (B) 35
(D) 34

47. In triangle ABC, $\angle C = 5\angle A$ and $\angle B = 3\angle A$. The triangle is, by its angles:

- (A) Acute
(C) Right-angled
- (B) Obtuse
(D) Equiangular

48. An exterior angle of a triangle is 105° and the interior angle ADJACENT to it is split by an angle bisector. The interior angle adjacent to the exterior angle is:

- (A) 75°
(C) 37.5°
- (B) 105°
(D) 52.5°

49. In triangle ABC, $\angle A = (4x)^\circ$, $\angle B = (3x)^\circ$ and $\angle C = (2x)^\circ$. Side a is opposite $\angle A$, side b opposite $\angle B$, side c opposite $\angle C$. Which side is the longest?

- (A) Side a (opposite $\angle A$)
(C) Side c (opposite $\angle C$)
- (B) Side b (opposite $\angle B$)
(D) All sides equal

50. In triangle ABC, $\angle A = 60^\circ$, and the bisectors of $\angle B$ and $\angle C$ meet at I. Then $\angle BIC$ equals:

- (A) 60°
(C) 150°
- (B) 120°
(D) 90°

51. A triangle has angles 30° , 30° and 120° . Two altitudes are drawn from the 30° vertices. These altitudes:

- (A) Fall inside the triangle
(C) Are equal to the median from the 120° vertex
- (B) Coincide with two sides of the triangle
(D) Fall outside the triangle (their feet are on extensions of the opposite sides)

52. In triangle ABC, side BC = 12 cm and the triangle is isosceles with AB = AC. If the perimeter is 32 cm and each equal side is a whole number, what is the length of each equal side?

- (A) 6 cm
(C) 12 cm
- (B) 10 cm
(D) 20 cm

53. In triangle ABC, $\angle B = 80^\circ$, and the exterior angle at A is 145° . Find $\angle C$.

- (A) 35°
(C) 80°
- (B) 100°
(D) 65°

54. Three rods of length 6 cm, 6 cm and 11 cm are joined. A fourth choice replaces the 11 cm rod with a rod of length L so a valid triangle forms with the two 6 cm rods. The largest whole-number L is:

- (A) 12 cm
(C) 11 cm
- (B) 6 cm
(D) 10 cm

55. In triangle PQR, the bisector of $\angle P$, the median from Q, and the altitude from R are all drawn. Which one of these MUST be perpendicular to the side it meets?

- (A) The altitude from R
(C) The median from Q
- (B) The bisector of $\angle P$
(D) All three of them

56. A triangle has angles in the ratio 1 : 2 : 6. By angles, the triangle is:

- (A) Acute
- (B) Obtuse
- (C) Right-angled
- (D) Equiangular

57. The sides of a triangle are three consecutive EVEN whole numbers and its perimeter is 30 cm. The triangle is, by sides:

- (A) Isosceles
- (B) Equilateral
- (C) Scalene
- (D) Cannot exist

58. In triangle ABC, $\angle A = 40^\circ$ and the bisector AD of $\angle A$ meets BC at D. In triangle ABD, $\angle ADB = 100^\circ$. Find $\angle B$.

- (A) 80°
- (B) 40°
- (C) 20°
- (D) 60°

59. Two exterior angles of a triangle, taken at two different vertices, are 130° and 112° . The third interior angle (at the remaining vertex) is:

- (A) 118°
- (B) 62°
- (C) 50°
- (D) 68°

60. In an isosceles triangle ABC with $AB = AC$, the perpendicular from A meets BC at D. If $BD = 7$ cm, what is the length of BC?

- (A) 7 cm
- (B) 14 cm
- (C) 21 cm
- (D) 3.5 cm

Solutions — Mathematics (Class 7), Chapter 7: A Tale of Three Intersecting Lines — Paper 3

Paper 3 — (progressively harder)

Marking: +4 correct, -1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	B	21	C	31	D	41	C	51	D
2	C	12	B	22	C	32	A	42	D	52	B
3	C	13	C	23	B	33	A	43	A	53	D
4	D	14	D	24	A	34	C	44	A	54	C
5	A	15	D	25	A	35	D	45	B	55	A
6	B	16	C	26	B	36	D	46	D	56	B
7	D	17	A	27	D	37	A	47	B	57	C
8	D	18	C	28	C	38	C	48	A	58	D
9	B	19	D	29	B	39	A	49	A	59	B
10	A	20	C	30	C	40	A	50	B	60	B

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (B) The angles sum to 180° , and $3 + 5 + 7 = 15$ parts, so one part is $180/15 = 12^\circ$. The largest interior angle is 7 parts = 84° . An exterior angle is the supplement of its adjacent interior angle: $180^\circ - 84^\circ = 96^\circ$. Reading 84° confuses the interior angle with its exterior partner; 108° supplements the 5-part (60°) wrongly; 72° supplements a value that does not occur.

2. (C) The exterior angle equals the sum of the two remote interior angles: $\angle P + \angle Q = 124^\circ$. With $\angle Q = \angle P - 26^\circ$, we get $2\angle P - 26^\circ = 124^\circ$, so $2\angle P = 150^\circ$ and $\angle P = 75^\circ$ (then $\angle Q = 49^\circ$). Picking 49° gives $\angle Q$ instead; 62° just halves 124° ignoring the difference; 98° wrongly treats 124° as adjacent to $\angle P$.

3. (C) The third side x satisfies $13 - 6 < x < 13 + 6$, i.e. $7 < x < 19$. The whole numbers strictly between are 8, 9, ..., 18, which is $18 - 8 + 1 = 11$. Counting 12 wrongly includes 7 or 19; 13 counts both endpoints; 7 stops short at 14.

4. (D) The two base angles are equal and together with the 44° vertex angle sum to 180° , so each is $(180^\circ - 44^\circ)/2 = 136^\circ/2 = 68^\circ$. The bisector detail does not affect the base angles. Choosing 44° repeats the vertex angle; 88° doubles a base angle; 46° comes from a faulty $(180-88)/2$ step.

5. (A) The angles sum to 180° : $(2x+10) + (3x-20) + (x+40) = 6x + 30 = 180$, so $6x = 150$ and $x = 25^\circ$. The angles are 60° , 55° and 65° , all below 90° , so the triangle is acute. It is not right-angled (no 90°), not obtuse (no angle over 90°), and not equiangular (the three differ).

6. (B) Three equal sides make a triangle equilateral, and an equilateral triangle has three 60° angles, so it is acute. The second has two equal angles (40° , 40°), so two equal sides — it is isosceles — and its 100° angle makes it obtuse. The first cannot be right (no 90°) or scalene (sides are equal); the second is not acute ($100^\circ > 90^\circ$) or right (no 90°).

7. (D) When the altitude and median from the same vertex coincide, that line is both perpendicular to BC and bisects it, so the two halves of the triangle are mirror images, forcing $AB = AC$. This need not be equilateral (only two sides are forced equal), need not be right-angled at A, and says nothing about BC being longest.

8. (D) In any triangle the largest angle lies opposite the longest side. The longest side is $BC = 9$ cm, and the angle opposite BC is $\angle A$ (the vertex not on BC). So $\angle A$ is the largest. $\angle B$ faces $CA = 5$ cm (shortest, so smallest), $\angle C$ faces $AB = 7$ cm (middle); they are not equal since the sides differ.

9. (B) Let $\angle B = b$. Then $\angle A = b + 18^\circ$ and $\angle C = 2b$. The sum is $(b + 18) + b + 2b = 4b + 18 = 180$, so $4b = 162$ and $b = 40.5^\circ$. Then $\angle C = 2b = 81^\circ$. Picking 40.5° reports $\angle B$; 60° assumes equal split; 63° is $\angle A$ wrongly named.

10. (A) The third angle is $180^\circ - 35^\circ - 85^\circ = 60^\circ$. Larger angles face longer sides, so ordering sides by their opposite angles $35^\circ < 60^\circ < 85^\circ$ gives shortest to longest exactly. The reverse order is backwards; sides are unequal since the angles differ; the side opposite 85° (not 60°) is the longest.

11. (B) The exterior angle at B equals the sum of the two remote interior angles: $\angle CBE = \angle A + \angle C$. So $3\angle C = 50^\circ + \angle C$, giving $2\angle C = 50^\circ$ and $\angle C = 25^\circ$. Then $\angle CBE = 75^\circ$. Choosing 75° reports the exterior angle; 50° just copies $\angle A$; 37.5° halves 75° incorrectly.

12. (B) The third side x satisfies $15 - 8 < x < 15 + 8$, i.e. $7 < x < 23$, so the smallest whole-number value is 8. Perimeter $= 8 + 15 + 8 = 31$ cm. Using 7 (which fails the strict inequality) gives 30; 23 wrongly adds only the two given sides; 38 uses the largest allowed value 15 instead of the smallest.

13. (C) Let the third angle be t ; each equal angle is $t + 15$, and there are two of them. Angle sum: $(t + 15) + (t + 15) + t = 3t + 30 = 180$, so $3t = 150$ and $t = 50^\circ$. Each equal angle is $t +$

$15 = 65^\circ$. Choosing 50° reports the third angle; 70° adds 20 instead of 15; 55° comes from a wrong $3t = 165$ slip.

14. (D) Since $\angle C = 80^\circ$, the other two sum to $\angle A + \angle B = 100^\circ$. With $\angle A : \angle B = 2 : 3$, the parts total 5, so one part is $100/5 = 20^\circ$. Then $\angle B = 3$ parts $= 60^\circ$ (and $\angle A = 40^\circ$). Choosing 40° reports $\angle A$; 50° splits 100° evenly; 100° is the combined sum, not $\angle B$ alone.

15. (D) In a right triangle the two legs are themselves the altitudes from the acute-angle vertices, and the third altitude from the right-angle vertex falls inside; all three altitudes pass through the right-angle vertex, so that is their common point. They do not meet at a midpoint, and for a right (or acute) triangle the meeting point is not outside.

16. (C) Since $AB = AC$, the base angles are equal: $\angle B = \angle C = 64^\circ$, so the vertex angle $\angle A = 180^\circ - 64^\circ - 64^\circ = 52^\circ$. The bisector from A splits $\angle A$ in half, so $\angle BAD = 52^\circ/2 = 26^\circ$. Choosing 52° gives the whole vertex angle; 32° halves 64° ; 64° copies a base angle.

17. (A) The triangle inequality needs the two smaller sides to sum to more than the largest: $2k + 3k = 5k$, which exceeds $4k$ for every positive k . So the set always works regardless of k 's exact value. The inequality does not depend on k being whole or greater than 1, and it never fails for positive k .

18. (C) Interior $\angle A = 180^\circ - 140^\circ = 40^\circ$ and interior $\angle B = 180^\circ - 110^\circ = 70^\circ$. Then $\angle C = 180^\circ - 40^\circ - 70^\circ = 70^\circ$. Alternatively the three exterior angles sum to 360° , so the exterior angle at C is $360^\circ - 140^\circ - 110^\circ = 110^\circ$, making interior $\angle C = 70^\circ$. Choosing 110° reports C's exterior angle; 60° and 40° come from arithmetic slips.

19. (D) The other two angles add to $180^\circ - 96^\circ = 84^\circ$. Since their total is under 90° , each must be below 90° , so both are acute. Neither can be 90° (that alone would exceed 84°), neither can be obtuse for the same reason, and there is no information forcing them to be equal.

20. (C) With two equal sides 10 and 10, the third side x satisfies $|10 - 10| < x < 10 + 10$, i.e. $0 < x < 20$. The whole numbers are 1, 2, ..., 19, giving 19 values. Counting 20 includes $x = 20$ (degenerate); 10 only counts up to 10; 21 includes both 0 and 20.

21. (C) Sum: $(x+30) + 2x + (x-10) = 4x + 20 = 180$, so $4x = 160$ and $x = 40^\circ$. The angles are 70° , 80° and 30° ; the largest is $\angle B = 2x = 80^\circ$. Choosing 70° gives $\angle A$; 60° comes from a wrong $x = 30$; 30° gives the smallest angle $\angle C$.

22. (C) The exterior angle at B is the supplement of the base angle $\angle B$, so $\angle B = 180^\circ - 115^\circ = 65^\circ$. Since $AB = AC$, $\angle C = \angle B = 65^\circ$, hence $\angle A = 180^\circ - 65^\circ - 65^\circ = 50^\circ$. Choosing 65° reports a base angle; 115° just repeats the exterior angle; 80° wrongly uses $180 - 2(50)$.

23. (B) A triangle is isosceles when at least two angles are equal. (45,45,90) has two 45° , (70,70,40) has two 70° , and (60,60,60) has all equal (equilateral, which is also isosceles)

— that is 3. The triple (100,50,30) has three different angles, so it is scalene. So exactly 3 qualify, not all 4 and not just 1 or 2.

24. (A) The exterior angle at C (125°) equals the sum of the two remote interior angles $\angle A + \angle B$. So $\angle A = 125^\circ - 65^\circ = 60^\circ$. Choosing 55° subtracts wrongly; 65° just copies $\angle B$; 115° treats 125° as supplementary to $\angle A$ instead of using the remote-angle rule.

25. (A) The third side x must satisfy $12 - 7 < x < 12 + 7$, i.e. $5 < x < 19$. The value 5 cm fails because $7 + 5 = 12$ only equals (does not exceed) the third side, so the points would be collinear. The values 8, 15 and 18 all lie strictly inside the range and are valid.

26. (B) A median joins a vertex to the midpoint of the opposite side, so $QM = MR$; the two triangles PQM and PMR share the same height from P and have equal bases, giving equal areas. A median is perpendicular only in special (isosceles) cases, it bisects the angle only in special cases, and there is no rule making it half of QR.

27. (D) The third side x satisfies $14 - 9 < x < 14 + 9$, i.e. $5 < x < 23$. The even whole numbers strictly between 5 and 23 are 6, 8, 10, 12, 14, 16, 18, 20, 22 — exactly 9 of them. Choosing 8 or 10 miscounts the list; 17 is the total number of all (not just even) whole-number values, ignoring the even restriction.

28. (C) Equal angles face equal sides. Since $\angle B = \angle C$, the sides opposite them are equal, so the side opposite $\angle C$ equals the side opposite $\angle B = 10$ cm. Choosing 6 cm uses the side opposite $\angle A$; 8 cm averages the two; 16 cm adds them.

29. (B) Exterior angles sum to 360° : $2y + 3y + 4y = 9y = 360$, so $y = 40^\circ$. The exterior angles are 80° , 120° and 160° . The largest exterior angle (160°) sits at the smallest interior angle, which is $180^\circ - 160^\circ = 20^\circ$. Choosing 40° gives y ; 80° gives the smallest exterior angle; $60^\circ = 180 - 120$ is the middle interior angle.

30. (C) In a right triangle the altitude from the right-angle vertex drops perpendicularly onto the hypotenuse and its foot lies inside that side, strictly between the two endpoints. It does not reach a vertex, does not fall outside the triangle, and is shorter than the hypotenuse, not equal to it.

31. (D) $\angle B + \angle C = 180^\circ - 40^\circ = 140^\circ$, so the half-angles sum to 70° . In triangle BIC the angle at I is 180° minus the two half-angles: $\angle BIC = 180^\circ - 70^\circ = 110^\circ$. (Equivalently $90^\circ + \text{half of } \angle A = 90^\circ + 20^\circ = 110^\circ$.) Choosing 70° forgets to subtract from 180° ; 140° uses the full angles; 90° drops the $\angle A/2$ term.

32. (A) The three angles are in arithmetic progression with common difference 20° : they are $\angle B + 20^\circ$, $\angle B$, $\angle B - 20^\circ$. Their sum is $3\angle B = 180^\circ$, so $\angle B = 60^\circ$. (The angles are 80° , 60° , 40° .) Choosing 40° gives $\angle C$; 80° gives $\angle A$; 50° comes from a faulty split.

33. (A) Two equal sides (5, 5) make it isosceles. To check the angles, the base 6 is shorter than the sum-square comparison would need for a right or obtuse angle; in fact with sides 5,5,6 the largest angle is opposite 6 and is acute (a 5-5-6 triangle is close to a 5-5-5 equilateral one, whose angles are all 60° , and lengthening one side slightly keeps that angle below 90°). It is not equilateral (sides differ) and not scalene (two sides equal).

34. (C) Let $\angle C = c$. Then $\angle A = 2c$ and $\angle B = 2c - 20$. Sum: $2c + (2c - 20) + c = 5c - 20 = 180$, so $5c = 200$ and $c = 40^\circ$. (Then $\angle A = 80^\circ$ and $\angle B = 60^\circ$.) Choosing 80° reports $\angle A$; 60° reports $\angle B$; 50° comes from a wrong $4c = 200$ slip.

35. (D) A line through A parallel to BC creates alternate angles equal to $\angle B$ and $\angle C$ on the two sides, so those two angles are 55° and 65° , summing to 120° . This is exactly $180^\circ - \angle A$, consistent with $\angle A = 60^\circ$. Choosing 60° gives $\angle A$ itself; 110° doubles 55° ; 130° doubles 65° .

36. (D) The median's foot M is the midpoint of BC; the altitude's foot H is where the perpendicular from A meets BC. They coincide only when $AB = AC$, i.e. when $\angle B = \angle C$. Here $\angle B = 40^\circ \neq 70^\circ = \angle C$, so $AB \neq AC$ and $M \neq H$. H is not the midpoint, and M is not the foot of the perpendicular.

37. (A) For an isosceles triangle the third side equals one of the given sides, so it is 16 or 9 (giving sides 16,16,9 or 16,9,9). Check validity: 16,16,9 works ($16+9>16$); 16,9,9 works ($9+9=18>16$). Both are valid, so there are 2 values. Saying 1 misses one case; 3 invents an extra; 0 ignores both.

38. (C) The exterior angle at C equals $\angle A + \angle B = 150^\circ$. With $\angle A = 4\angle B$, we get $4\angle B + \angle B = 5\angle B = 150^\circ$, so $\angle B = 30^\circ$ and $\angle A = 120^\circ$. Choosing 30° reports $\angle B$; 96° and 100° are faulty splits of 150° .

39. (A) Parts: $3 + 4 + 5 = 12$, so one part is $36/12 = 3$ cm, giving sides 9, 12 and 15 cm; the longest is 15 cm. A 3-4-5 (here 9-12-15) triangle has $9^2 + 12^2 = 81 + 144 = 225 = 15^2$, a Pythagorean relation, so it is right-angled. Choosing 'acute' misreads the angle; 12 cm picks the middle side; 20 cm misuses the parts.

40. (A) $\angle B + \angle C = 180^\circ - 70^\circ = 110^\circ$. With $\angle B = 3\angle C$, we get $3\angle C + \angle C = 4\angle C = 110^\circ$, so $\angle C = 27.5^\circ$ and $\angle B = 82.5^\circ$. Choosing 27.5° reports $\angle C$; 55° splits 110° evenly; 110° is the combined sum, not $\angle B$.

41. (C) Let each base angle be b ; there are two of them and the vertex angle is $4b$. Sum: $b + b + 4b = 6b = 180^\circ$, so $b = 30^\circ$ and the vertex angle is $4b = 120^\circ$. Choosing 30° gives a base angle; 60° and 90° come from wrong relationships.

42. (D) Validity needs $20 - 13 < x < 20 + 13$, i.e. $7 < x < 33$, which every value from 13 to 20 already satisfies. The whole numbers from 13 to 20 inclusive are 13, 14, ..., 20, that is $20 - 13 + 1 = 8$ values. Choosing 7 forgets an endpoint; 6 undercounts; 33 confuses the count with the upper bound.

43. (A) The bisector of $\angle A$ splits it into 25° on each side, so in triangle ABD the angle $\angle BAD = 25^\circ$. With $\angle ABD = 60^\circ$, the third angle $\angle ADB = 180^\circ - 25^\circ - 60^\circ = 95^\circ$. Choosing 85° forgets to halve $\angle A$; 70° uses the full 50° ; 65° is an unrelated subtraction.

44. (A) All three angles different means all three sides differ, so the triangle is scalene. The largest angle 95° exceeds 90° , so it is obtuse. It is not acute (an angle over 90° exists), not isosceles (no two angles equal), and not right (no 90° angle).

45. (B) An altitude's foot falls outside the opposite side only when the angle at the near vertex is obtuse; here the foot is beyond B, so $\angle B$ is obtuse. It is not $\angle A$ that is obtuse (that altitude is from A and still drops onto BC's line), it is not a right angle (then the foot would be at B), and an equilateral triangle has all feet inside.

46. (D) The third side x satisfies $25 - 18 < x < 25 + 18$, i.e. $7 < x < 43$. The largest whole-number value is 42 and the smallest is 8, so the difference is $42 - 8 = 34$. Choosing 43 subtracts the loose bounds; 35 uses $43 - 8$; 33 uses $42 - 9$ or similar slips.

47. (B) Let $\angle A = a$; then $\angle B = 3a$ and $\angle C = 5a$. Sum: $a + 3a + 5a = 9a = 180^\circ$, so $a = 20^\circ$, giving angles 20° , 60° and 100° . The largest angle, $\angle C = 100^\circ$, exceeds 90° , so the triangle is obtuse. It is not acute (an angle over 90° exists), not right (no 90°), and not equiangular (the angles differ).

48. (A) An exterior angle and its adjacent interior angle are supplementary, so the adjacent interior angle is $180^\circ - 105^\circ = 75^\circ$. The bisector detail does not change this value. Choosing 105° copies the exterior angle; 37.5° halves 75° (the bisected halves, not the whole); 52.5° halves 105° .

49. (A) Sum: $4x + 3x + 2x = 9x = 180^\circ$, so $x = 20^\circ$ and the angles are 80° , 60° and 40° . The longest side lies opposite the largest angle; $\angle A = 80^\circ$ is largest, so side a is longest. Side b faces 60° (middle) and side c faces 40° (shortest); the sides are not equal since the angles differ.

50. (B) $\angle B + \angle C = 180^\circ - 60^\circ = 120^\circ$, so the half-angles sum to 60° . In triangle BIC, $\angle BIC = 180^\circ - 60^\circ = 120^\circ$ (equivalently $90^\circ + \angle A/2 = 90^\circ + 30^\circ = 120^\circ$). Choosing 60° forgets the 180° subtraction; 150° miscomputes the half-sum; 90° drops the $\angle A/2$ term.

51. (D) Because the triangle has an obtuse angle (120°), the altitudes from the two acute (30°) vertices land outside the triangle, on the extensions of the sides adjacent to the obtuse angle. They do not fall inside (that happens in acute triangles), they coincide with sides only at a right angle, and there is no rule equating them to that median.

52. (B) The two equal sides together measure $32 - 12 = 20$ cm, so each equal side is $20/2 = 10$ cm. Check validity: $10 + 10 = 20 > 12$, valid. Choosing 6 cm wrongly subtracts 12 from 20 and halves; 12 cm copies the base; 20 cm gives the total of the two equal sides, not one.

53. (D) The exterior angle at A equals the sum of the remote interior angles $\angle B + \angle C = 145^\circ$. So $\angle C = 145^\circ - 80^\circ = 65^\circ$. Choosing 35° subtracts 145 from 180; 100° supplements 80° ; 80° copies $\angle B$.

54. (C) With two sides 6 and 6, the third side L satisfies $0 < L < 12$, so the largest whole number is 11. (Indeed $6 + 6 = 12$ must exceed L, so $L = 11$ works but 12 fails.) Choosing 12 cm violates the strict inequality; 6 cm is too small to be the maximum; 10 cm is allowed but not the largest.

55. (A) By definition an altitude is the perpendicular segment from a vertex to the opposite side (or its extension), so the altitude from R is perpendicular to PQ. An angle bisector splits the vertex angle, not necessarily meeting the side at a right angle; a median hits the midpoint, not necessarily perpendicularly; so not all three are perpendicular.

56. (B) Parts: $1 + 2 + 6 = 9$, so one part is $180/9 = 20^\circ$. The angles are 20° , 40° and 120° . The 120° angle exceeds 90° , so the triangle is obtuse. It is not acute (an angle over 90° exists), not right (no 90°), and not equiangular (angles differ).

57. (C) Three consecutive even numbers summing to 30 are 8, 10 and 12 ($8+10+12 = 30$). They satisfy $8 + 10 = 18 > 12$, so a valid triangle forms, and since all three sides differ it is scalene. It is not isosceles or equilateral (no equal sides), and it does exist.

58. (D) The bisector makes $\angle BAD = 40^\circ/2 = 20^\circ$. In triangle ABD the angles sum to 180° , so $\angle B = 180^\circ - 100^\circ - 20^\circ = 60^\circ$. Choosing 80° forgets to halve $\angle A$; 40° copies $\angle A$; 20° gives $\angle BAD$ instead of $\angle B$.

59. (B) Interior angle at the first vertex = $180^\circ - 130^\circ = 50^\circ$; at the second = $180^\circ - 112^\circ = 68^\circ$. The third interior angle = $180^\circ - 50^\circ - 68^\circ = 62^\circ$. Choosing 118° gives the third exterior angle ($360 - 130 - 112 = 118$); 50° and 68° are the other two interior angles, not the third.

60. (B) In an isosceles triangle the perpendicular (altitude) from the apex A to the base BC also bisects the base, so D is the midpoint and $DC = BD = 7$ cm. Therefore $BC = BD + DC = 7 + 7 = 14$ cm. Choosing 7 cm gives only half; 21 cm trebles; 3.5 cm halves the wrong way.